Support bounds without a parity hypothesis

Human Compression project: a component of the L019 recovery

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ be a transitive string action by involutions, with connected noncommutation diagram and all maximal omissions intransitive. Let a unique eligible i-edge join $a \in A$ to $b \in B$, the two omission orbits. Assume $|A|, |B| \geq 2$ and $L = G_{<i}$ fixes B pointwise. Write $R = G_{>i}$. These two groups commute, their product is transitive on A, and R is transitive on B.

We acquire the exact unique-crossing and fixed-prefix arguments from the earlier work. In particular, if $l > i$ acts on either side, there is a simple path from that side's crossing endpoint to an l-edge, using only earlier indices different from i. The path with its last edge has at least $l - i$ edges; its prefix, excluding that last edge, is fixed pointwise by $G_{>l}$. Every intermediate index $i + 1, \dots, l - 1$ occurs. These assertions are generic and use no evenness or intersection property.

Lemma 1 (The commuting-orbit counts)**.** *Suppose $i > 0$ and L is intransitive on A. Its orbits in A have a common size $k \geq i + 1$. Put $\beta = a^L$ and $\tau = \rho_{i+1}$. Then τ exchanges β with a distinct orbit β'. All marks of index at least $i + 2$ fix β pointwise, and those of index at least $i + 3$ fix β' pointwise. If ρ_h acts nontrivially on A, where $d = h - i \geq 2$, there are at least d such orbits. If the first active orbit reached by the earlier marks is moved to another orbit by ρ_h, there are at least $d + 1$.*

Proof. The group R transports the L-orbits equivariantly and transitively, so their sizes agree. A subfamily of the L marks transitive on one orbit is transitive on all of them. Together with R it is transitive on both A, B. The acquired minimal-transitivity lemma therefore requires every L mark in this subfamily. Its orbit forest gives $i \leq k - 1$.

All indices at least $i + 2$ fix a by the unique crossing and commute with L, hence fix β pointwise. Therefore τ must send β elsewhere, or it would be invariant under G_i. This gives β'. Commutation with τ then makes indices at least $i + 3$ fix β' pointwise.

Consider the L-orbit graph using indices strictly between i and h, starting at β. Some reachable orbit has nontrivial ρ_h-action. Otherwise ρ_h would fix their union pointwise; every higher mark does so too, since it fixes β and commutes with all the path marks. Their union would be G_i-invariant, hence all of A, contradicting the assumed action of ρ_h. Choose a shortest orbit path to such an orbit. Every intermediate index occurs in its word: if one were missing, commute the higher-index letters to the beginning. They fix a, and the remaining lower-index word commutes with ρ_h and starts at a point fixed by it. This would make its endpoint fixed by ρ_h. At an active invariant orbit this is impossible, since a nonidentity centralizer of a transitive L-action has no fixed point; if that orbit is moved, it is impossible immediately. The path has at least $d-1$ edges and d distinct orbits. Every earlier orbit is fixed pointwise by ρ_h, by minimality. If the last one is moved, its image is a further orbit, proving the last count. □

Lemma 2 (An internal action at distance two)**.** *Under Lemma 1, suppose $\sigma = \rho_{i+2}$ preserves β' and is nontrivial there. Then*

$$A = \beta \cup \beta', \qquad k \geq 4, \qquad i \leq k/2 - 1.$$

Proof. Every later generator fixes $\beta \cup \beta'$ pointwise; L, τ, σ preserve its union. Transitivity on A therefore gives the asserted union. On β', σ centralizes the transitive L-action, so is fixed-point-free. Its $k/2$ pairs are L-invariant, and σ^τ supplies the corresponding pairs in β. Choose

a subfamily M of the L marks transitive on these pairs, with $|M| \le k/2-1$. The group $\langle M, \sigma^\tau \rangle$ is transitive on β, and $\langle M, \sigma \rangle$ on β'. Together with τ they are transitive on A. The whole R remains transitive on B. Minimal transitivity at the crossing forces M to contain all L marks. This proves $i \le k/2-1$; since $i > 0$, $k \ge 4$. □

Lemma 3 (A transitive first side)**.** *If $i > 0$, L is transitive on A, and a mark after i acts nontrivially on A, that mark has index $i+1$. Moreover $|A| = 2m$ with $i \le m-1$.*

Proof. Every mark of index at least $i+2$ fixes a and centralizes the transitive L-action, so fixes A pointwise. Thus the nontrivial action belongs to $\tau = \rho_{i+1}$, whose restriction is fixed-point-free. Its m pairs are L-invariant. Choose at most $m-1$ of the L marks transitive on the pairs. Together with τ these are transitive on A, and the whole R is transitive on B. Minimal transitivity forces all i marks to be present, as claimed. □

Lemma 4 (The outside side alone)**.** *If $h > i$ and ρ_h acts nontrivially on A, then*

$$|A| \ge i + h + 1.$$

For $i = 0$ no hypothesis that $G_{<i}$ be nontrivial is needed.

Proof. For $i = 0$ the acquired path has at least $h+1$ vertices in A. For $i > 0$ and transitive L, Lemma 3 gives $h = i+1$ and $|A| \ge 2i+2$. Otherwise use Lemma 1 and set $d = h - i$. For $d = 1$, $|A| \ge 2k \ge 2i+2$. For $d = 2$, the first active orbit is β'. If it is moved, $|A| \ge 3k \ge 3i+3$ is sufficient. If it is preserved nontrivially, Lemma 2 gives $|A| = 2k \ge 4i+4$. It cannot be fixed pointwise, since then $\beta \cup \beta'$ would be invariant under every mark of G_i, and ρ_h would be trivial on A. For $d \ge 3$,

$$|A| \ge d(i+1) \ge 2i + d + 1 = i + h + 1,$$

where $i(d-2) \ge 1$. This includes the possible equality $i = 1, d = 3$; no parity argument is used to discard it. □

Lemma 5 (A last mark acting on the first side)**.** *If $i > 0$ and ρ_{r-1} acts nontrivially on A, then $n \ge 2r$. If $i = 0$ and the last mark acts on both sides, the same conclusion holds.*

Proof. For $i = 0$, take the two disjoint acquired paths, each with r vertices. Now suppose $i > 0$. If L is transitive on A, Lemma 3 gives $r - 1 = i + 1$. On B, R is the single involution τ, so $|B| = 2$. Thus $n = 2m + 2 \ge 2i + 4 = 2r$.

Suppose L is intransitive. First let the last mark fix B pointwise. Its image on the L-orbits in A cannot belong to the group generated by the other R images: otherwise those images would remain transitive on the orbit set, and L with the remaining R marks would be transitive on both A, B. This contradicts minimal transitivity. In particular its block image is nontrivial. Put $d = r - 1 - i \ge 2$, since τ moves b. A shortest block path from β to a block moved by the last mark uses each intermediate index, by the same no-gap argument as Lemma 1. It visits at least d distinct blocks; the last mark sends its final block to one more. Hence

$$n \ge (i+1)(d+1) + 2 = 2r + (i-1)(d-1) \ge 2r.$$

Finally suppose the last mark acts on both sides. If $d = 1$, $A = \beta \cup \beta'$ and $|B| = 2$, giving $n = 2k + 2 \ge 2i + 4 = 2r$. If $d = 2$, the active orbit β' is either moved, giving $|A| \ge 3k$, or preserved nontrivially, giving Lemma 2. In the first case the path in B gives $|B| \ge 3$, so $n \ge 3i + 6 \ge 2r$. In the second, $n \ge 2k + 3$ and $r = i + 3 \le k/2 + 2$, which again imply $n \ge 2r$ because $k \ge 4$. For $d \ge 3$, the orbit count and the path in B give

$$n \ge d(k+1) + 1 \ge d(i+2) + 1 = 2r + i(d-2) - 1 \ge 2r.$$

These cases exhaust the last mark's possible action. □

Theorem 6 (The head support bound)**.** *Assume $2r > n + 2$ and at least one mark after i is shared by both sides. Let $h > i$ be the largest shared index. Then $h < r - 1$. If $i = 0$, interchange A, B when necessary so that the final tail acts on B. For $i > 0$ this is already its side. With $X = \mathrm{supp}(G_{>h})$,*

$$X \subseteq B \setminus \{b\}, \qquad |\operatorname{Fix}(G_{>h})| \geq 2h + 1.$$

Proof. Lemma 5 excludes $h = r - 1$. Every later mark is nonidentity and acts on exactly one side. Connectedness of the noncommutation diagram forces the same side for all of them: a change of sides would make adjacent marks commute. For $i > 0$, Lemma 5 excludes A as this side. The tail therefore fixes all of A and also fixes b by the unique crossing.

For $i = 0$, the path in A including its last h-edge has at least $h + 1$ vertices, all fixed by the tail. The path prefix in B has at least h vertices, also fixed. These disjoint sets give $2h + 1$ fixed points.

Let $i > 0$. If L is transitive, then $h = i + 1$ and $|A| + 1 \geq 2i + 3 = 2h + 1$ by Lemma 3. Otherwise put $d = h - i$. The fixed prefix in B has at least d vertices, so the fixed set has size at least $|A| + d$. If $d = 1$, this is at least $2k + 1 \geq 2h + 1$. If $d = 2$ and the active orbit is moved, it is at least $3k + 2 \geq 3i + 5 \geq 2h + 1$. In the internal case it is at least $2k + 2$, whereas $i \leq k/2 - 1$ and $k \geq 4$ imply $2h + 1 = 2i + 5 \leq k + 3 \leq 2k + 2$. For $d \geq 3$, it is at least

$$d(k + 1) \geq d(i + 2) = 2h + 1 + i(d - 2) - 1 \geq 2h + 1.$$

The support inclusion follows from the already proved fixation of A and b. □

Attribution and boundary. The source is Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2, Propositions 5.8–5.12. Its path and commuting-orbit strategy, and the complete generic path proofs already acquired for L044, remain attributed. The preceding L044 orbit-distance, transitive-side and internal-block arguments are reworked here with minimal transitivity replacing faithful quotient rank bounds. Every change in hypothesis is proved: evenness and the alternating small-degree assumptions are not used, and the old parity exclusions of equality are not claimed. The support X is on the side *moved* by the final tail; the source's contrary orbit wording is not retained.

This manuscript establishes the exact weak inequalities needed in the strict L019 range. It does not claim the stronger equality classifications of the source or the alternating manuscripts, or yet establish existence of a perfect split. No new census, whole-original, conjecture, Lean or measurement credit is asserted.